

Cover letter — *Journal of Economic Growth*

Editor
Journal of Economic Growth

Dear Editor,

I am pleased to submit “The Long Shadow of Seasonality: Climate Endowments, Agricultural Pathways, and Escape from the Malthusian Trap” for consideration at the *Journal of Economic Growth*.

The paper sits squarely in the journal’s flagship topic. Building on [matranga2024](#), who shows that intra-annual climate variability drove the Neolithic transition, and on the unified-growth framework of [galor2011unified](#), I ask whether the climate endowment that shaped where humans first farmed also shaped where they first escaped the Malthusian trap. The empirical strategy connects pre-industrial agricultural pathway selection (Stage 1) to the strength of the Malthusian feedback and the response to climate shocks (Stage 2), and tests both stages on a new country-level and sub-national monthly climate panel that reaches back to 1421 CE.

The principal contributions are empirical. *First*, I construct what I believe is the first country-level and sub-national monthly climate panel reaching back to 1421 CE, by area-weighting the ModE-RA paleo-reanalysis onto HYDE 3.5 sub-units and calibrating against ERA5 over the 1950–2008 overlap. The country-monthly correlation against country-level ERA5 is 0.971 in median and the year-to-year anomaly correlation is 0.938, sharp enough for the analyses I run. *Second*, I exploit HYDE’s sub-national grid (3,187 sub-units across 196 countries) for within-country identification of the storage-demand → agricultural-system channel. With country fixed effects absorbing all country-level institutional confounders, an extra climatically productive month per year is associated with 6.7 pp higher crop share at 1750 and 0.66 log-units higher population density—an effect $3.5\times$ the between-country one. *Third*, on a 1421–1950 within-country Malthusian regression with real annual climate forcing, I recover a within-paper signature of the demographic transition (density coefficient progressively weakening, statistically zero by 1900–1950) and an underappreciated finding: the sign on inter-annual temperature volatility flips between 1750–1900 ($+0.0045$, helpful or noisy) and 1900–1950 (-0.0125 , $p < 10^{-4}$, strongly harmful), consistent with monoculture replacing mixed pre-industrial systems and exposing modern agriculture to volatility risk.

I see the paper as a contribution to three lines of work this journal has shaped: the long-run consequences of climate endowments [andersen2017climate](#), [olsson2005](#), the structural transition from stagnation to growth [galor2002natural](#), [voigtlander2013how](#), and the data infrastructure for pre-industrial comparative economics [ashraf2011dynamics](#). The within-country identification opens a route for the field to move past cross-country growth regressions where it is feasible to do so.

I declare no conflicts of interest. The manuscript has not been previously published and is not under consideration elsewhere. A replication package with the ModE-RA aggregation code, the country and sub-national parquets, and all analysis scripts will be deposited on OSF upon acceptance.

Sincerely,

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Suggested referees (independent of the author): T.B. Andersen (Copenhagen, climate-growth); A. Matranga (NES, seasonality-agriculture); Q. Ashraf (Williams, long-run development); A. Vollrath (Toulouse, paleo-economic history); A. de Pleijt (Wageningen, Dutch agricultural history).